Practice Worksheet: Gerrymandering

Recall the 3 quantitative approaches to measuring gerrymandering discussed in lecture:
1. Mean - Median Score (Grofman, 1983)
2. Efficiency Gap (Stephanopolous and McGee, 2015)
3. Ensembles / Extreme Outliers

Use the following table to practice applying the first two techniques.

District	D Votes	R Votes	Result
1	62	38	D wins
2	55	45	D wins
3	44	56	R wins
4	47	53	R wins
5	46	54	R wins
Total	**254**	**246**	

Q1. Mean-Median Score

Calculate the mean-median score associated with this particular district breakdown of votes.

```
view(votes)

votes2 <- votes |>
    mutate(v_i = r_votes / (r_votes + d_votes)) |>
    summarize(mean_v = mean(v_i),
    med_v = median(v_i)) |>
        mutate(mean_v - med_v)

view(votes2)
# mean(v_i) -> what statewide Republican vote shares look like
# median(v_i) -> what middle district vote shares look like (the one that
decides who wins majority)
# If mean(v_i) - median(v_i) < 0: if median republican votes are higher than
mean, that means the median district is more republican than the overall
statewide average -> that means that Republicans would need more than 50% of
statewide votes to  win majoirty seats -> so this district map would favor
democrats -> *** median district more Republican than state average, so
Republicans would have need MORE votes to win majority, and Republicans
WASTING THEIR VOTES
# where v_i = republican vote share in each district
# If mean(v_i) - median(v_i) > 0: if median republican votes are lower than
```

```
mean, that means the median district is less republican than statewide average
-> so Republicans can win majority seats if they have less than 50% of total
votes statewide -> so that district map would favor Republicans -> *** median
district less Republican (MORE democratic) than state average, so democrats
would be WASTING THEIR VOTES
# so basically if v_i = republican vote share in a district: then negative
mean-median value -> favor democrat, positive mean-median value -> favor
republican
```

Q2. Efficiency Gap

Calculate the Efficiency Gap (EG) associated with this particular district breakdown of votes.

Tip: it may be helpful to write a function that takes a vector of votes across districts for party A and vector of votes for the party B and returns a vector of the number votes wasted by party A across districts.

```r
view(votes)
```

```r
eg_gap <- votes |>
    mutate(
        winner = if_else(d_votes > r_votes, "D", "R"),
        d_wasted = if_else(winner == "D", d_votes - 51, d_votes),
        r_wasted = if_else(winner == "R", r_votes - 51, r_votes)
    )

eg_gap |>
    summarize(
        total_d_wasted = sum(d_wasted),
        total_r_wasted = sum(r_wasted),
        total_votes = sum(d_votes + r_votes)
    ) |>
        mutate(eg_gap = (total_d_wasted - total_r_wasted) / total_votes)
```

```
# A tibble: 1 × 4
  total_d_wasted total_r_wasted total_votes eg_gap
           <dbl>          <dbl>       <dbl>  <dbl>
1            152             93         500  0.118
```

d_wasted_votes_total <- 152 r_wasted_votes_total <- 93 (152 - 93) / 500 = 0.118

This means Democrats wasted more votes than Republicans (152 votes), so would be an advantage for Republicans since the EG value is a positive one (0.118).

Also if you flip the EG equation to (total_r_wasted - total_d_wasted) / total_votes, then the meaning of the signs would be flipped, so if EG value is positive, then democrat advantage, if EG value is negative, then Republican advantage.